

Programme	B. Sc. Computer Science				
Course Title	System Software				
Type of Course	Elective				
Semester	VIII				
Academic Level	400 - 499				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	13. Introduction to Computer Science 14. Data Structures and Algorithms 15. Computer Organization and Architecture				
Course Summary	With an emphasis on the creation and use of system software, this course examines the ideas and methods of system programming. Compiler design, system calls, loaders and linkers, and debugging methods are among the topics covered.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define the key concepts in system programming, such as compilers, assemblers, linkers, and loaders. Identify the various stages in the compilation process and understand the purpose of each stage.	U	C	Instructor-created exams / Quiz
CO2	Master different types of system calls and their role in system programming. Master the principles of lexical and syntax analysis in the context of compiler design. Master various linking and loading schemes	Ap	P	Assignment / Seminar presentations/ Exams
CO3	Interpret and understand the process of debugging, including the use of debugging tools and techniques.	Ap	P	Seminar Presentation / Group Tutorial Work/ Viva Voce
CO4	Analyze the impact of different optimization techniques in the compilation process. Evaluate advantages and disadvantages of various linking and loading schemes.	U	C	Instructor-created exams / Home Assignments

CO5	Implement programs using system calls to perform various system-level tasks, such as file operations and process management	Ap	P	Writing assignments
CO6	Apply principles of compiler design to write a simple compiler using a programming language.	Ap	P	Case Study
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs
I	Introduction to System Programming & Assemblers (18 Marks)		14
	1	Introduction to System Programming- Goals of System Software, System Programs and Systems Programming	3
	2	Language Processors- Overview, Kinds of Language processors, language processing activities, program execution	4
	3	System Tables	1
	4	Assemblers- Elements of Assembly Language Programming	2
	5	Design of two pass assembler	4
II	Macros and Macro Processors (16 Marks)		10
	6	Introduction to macros and macro processors- macro definition and call, macro expression	4
	7	Nested macro calls	2
	8	Design of macro processor	4
III	Linkers and Loaders, Scanning and Parsing (16 Marks)		10
	9	Relocation and linking concepts	2
	10	Design of linkers	2
	11	Self locating program	2
	12	Loaders- absolute loader, relocating loader	4
IV	Compilers, System calls and libraries (20 Marks)		14
	13	Data structures used in compilers	1
	14	Phases of a compiler – Introduction	1
	15	Lexical Analysis (Scanning)	2
	16	Syntax Analysis (Parsing)	2
	17	Semantic Analysis	1
	18	Intermediate code generation	1
	19	Code optimisation- optimisation transformation, local optimisation, global optimisation, Code Generation	2
	20	Passes of Compiler	1
	21	System calls and their implementation	1
	22	Standard C library functions for system calls	2
V	Open Ended Module: Case Studies		12
	1	- Case studies of lexical and syntax analyzers: LEX and YACC. - System programs using system calls	

References

- D.M. Dhamdhare, Systems Programming and Operating Systems
- John J Donovan, Systems programming
- Jim Welsh and R M McKeag, Structured System Programming, Prentice Hall.

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	1	-	1	-	-						
CO 2	2	3	-	1	-	-						
CO 3	1	-	1	1	1	-						
CO 4	2	2	1	1	-	-						
CO 5	2	3	1	-	-	-						
CO 6	2	3	1	-	-	-						

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	End Semester Examinations
CO 1	✓		✓
CO 2	✓		✓
CO 3	✓		✓
CO 4		✓	✓
CO 5		✓	✓
CO 6		✓	